SalesGenie AI — Software Requirements Specification

Status: Review and sign-off candidate
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Prepared: 15 October 2026
Source: Project brief, milestone plan, execution/training guides, and the updated 12–13 October task rows

> This document is prepared for stakeholder review. It is not approved or signed on behalf of the team. Approval fields are intentionally blank.

1. Purpose

SalesGenie AI is an AI-assisted sales platform for managing prospects, preparing sales outreach, analyzing leads, summarizing sales conversations, and monitoring sales activity. It is intended to reduce repetitive research and administration while helping sales users prioritize and follow up with prospects.

This SRS records the agreed project scope and provides a basis for API design, implementation, and acceptance testing. Milestone 1 is the first implementation focus; the requirements below describe the full six-module project.

2. Project objectives

- Maintain a centralized, searchable lead and prospect database.
- Generate company and lead insights linked to the correct lead.
- Support personalized outreach and lead qualification.
- Summarize sales conversations and capture action items.
- Provide sales-pipeline and activity analytics.
- Expose backend functionality through a documented API and a Streamlit user interface.

3. Users and operating context

3.1 User

- **Sales user:** manages prospects, reviews insights, prepares outreach, records or summarizes conversations, and checks sales activity.

3.2 System components

- **User interface:** Streamlit dashboard.
- **API:** Python FastAPI service with Pydantic request/response validation.
- **Persistence:** SQLAlchemy ORM and Alembic migrations; SQLite for local development and PostgreSQL for production, following the selected training-guide baseline.
- **AI service:** OpenAI API, with the model selected through configuration. The specific model remains to be confirmed by the team.

The application must not be exposed to the public internet with real contact data until authentication, authorization, deployment security, and operational controls are implemented.

4. Scope

4.1 Included modules

1. **Lead Management & Prospect Database** — create, import, view, search, update, and track leads.

2. **Lead Intelligence & Company Analysis** — analyze company information and save retrievable insights against a lead.
3. **AI Outreach Generation** — draft personalized emails and sales messages from lead context.
4. **Lead Scoring & Recommendation Engine** — calculate a qualification/likelihood score and suggest next actions.
5. **Conversation Intelligence & CRM Integration** — summarize calls/meetings, extract actions, and support CRM synchronization.
6. **Dashboard & Sales Analytics** — present lead, pipeline, conversion, and outreach activity measures.

4.2 Milestone 1 focus

The first milestone is the lead-to-insight workflow:

1. Create or import a prospect.
2. Retrieve, search, and update the prospect.
3. Analyze company information.
4. Persist the resulting insights against the correct lead.
5. Display the lead and saved insights in the Leads screen.

CSV/manual import, duplicate identification, basic qualification information, test evidence, and a demo path are included in the milestone plan. The separate 85% scoring-accuracy target belongs to Milestone 2 and is not a Milestone 1 acceptance claim.

4.3 Excluded from the first milestone

- Production-grade authentication, role-based access control, and multi-tenancy.
- Live CRM synchronization or a commitment to a specific CRM vendor.
- Unattended or autonomous outreach delivery.
- A guarantee that generated company facts are accurate without supplied or verifiable sources.
- The Milestone 2 scoring accuracy target and later milestone outcomes.

5. Functional requirements

5.1 Lead management

- **FR-LEAD-001:** The system shall create a lead with a company name and optional industry, contact name, email, and phone.
- **FR-LEAD-002:** The system shall assign a unique lead ID and creation/update timestamps.
- **FR-LEAD-003:** The system shall list and retrieve leads, including paginated search/filter results.
- **FR-LEAD-004:** The system shall update lead details and lifecycle status.
- **FR-LEAD-005:** The system shall accept a CSV of leads and report created, duplicate, and invalid rows without silently discarding row errors.
- **FR-LEAD-006:** The system shall identify likely duplicates. Proposed matching: normalized email when present; otherwise normalized company name plus contact name. The team must confirm this rule before implementation.
- **FR-LEAD-007:** Invalid input and unknown lead IDs shall produce clear, user-visible errors and shall not crash the application.

Proposed lifecycle statuses for review: `New`, `Contacted`, `Qualified`, `Proposal`, `Won`, and `Lost`.

5.2 Company intelligence

- **FR-INTEL-001:** A user shall be able to request company analysis for an existing lead.

- **FR-INTEL-002:** An analysis shall be linked to the lead ID used for the request.
- **FR-INTEL-003:** The system shall save and retrieve analysis results after the request completes.
- **FR-INTEL-004:** An analysis shall return a concise summary, potential needs, opportunities, and qualification information where available.
- **FR-INTEL-005:** The UI shall show saved analysis alongside the corresponding lead.
- **FR-INTEL-006:** The system shall distinguish model-generated content from verified facts and shall not represent unsupported claims as confirmed facts.

5.3 Outreach generation

- **FR-OUT-001:** A user shall be able to request a draft email or sales message using a lead and selected context.
- **FR-OUT-002:** The system shall return a draft for user review; it shall not send messages automatically in the project scope described here.

5.4 Lead scoring and recommendations

- **FR-SCORE-001:** The system shall produce a lead score and explain the principal factors used.
- **FR-SCORE-002:** The system shall provide a next-best-action recommendation when sufficient lead/activity data is available.
- **FR-SCORE-003:** Scores shall be labeled as estimates, not guarantees of conversion.

5.5 Conversation intelligence and CRM integration

- **FR-CONV-001:** A user shall be able to submit a call/meeting transcript for summarization.
- **FR-CONV-002:** The system shall return a summary, key points, and action items.
- **FR-CRM-001:** The system shall support synchronization with an agreed CRM through documented integration contracts. The CRM vendor and synchronization scope remain open decisions.

5.6 Dashboard and analytics

- **FR-DASH-001:** The dashboard shall show total leads and lead counts by lifecycle status.
- **FR-DASH-002:** As relevant data becomes available, the dashboard shall show pipeline, conversion, and outreach activity measures.
- **FR-DASH-003:** Analytics shall be derived from persisted application data and shall not fabricate unavailable measures.

6. Data requirements

The initial logical data model is:

Entity	Purpose	Key relationships
Lead	Company/contact details and lifecycle status	One lead may have many insights, scores, outreach drafts, activities, and conversation summaries
CompanyInsight	Persisted company analysis and qualification information	Belongs to one lead
LeadScore	Score, rationale, and scoring timestamp	Belongs to one lead
OutreachDraft	Generated message, type, and creation timestamp	Belongs to one lead
Activity	Lead engagement history and event details	Belongs to one lead

Entity	Purpose	Key relationships
ConversationSummary	Summary, key points, and action items	Belongs to one lead when the conversation is associated with a lead

The team shall review the final column types, nullability, indexes, and migration strategy in the API/data contract before implementation. The initial lead fields are `lead_id`, `company_name`, `industry`, `contact_name`, `email`, `phone`, `lead_status`, `created_at`, and `updated_at`.

7. Non-functional requirements

- **NFR-REL-001:** At least 95% of the agreed Milestone 1 prospect test set shall be created/imported, retrieved, updated, and analyzed without an unhandled failure. The test report shall record numerator and denominator.
- **NFR-REL-002:** Missing IDs, invalid records, duplicate imports, and provider/database failures shall produce explicit errors; they shall not be reported as successful operations.
- **NFR-SEC-001:** API keys and populated environment files shall remain outside version control. Use a documented `.env.example` containing placeholders only.
- **NFR-SEC-002:** Demo and test runs shall use synthetic or approved sample records, not personal or confidential customer data.
- **NFR-SEC-003:** Local services shall bind to localhost by default. Production exposure requires a separately reviewed security design.
- **NFR-DATA-001:** A successful analysis shall persist with a valid lead foreign key and be retrievable after the request.
- **NFR-USE-001:** The user shall be able to complete the demo flow: Add Lead → retrieve lead → Analyze Company → view saved insights.
- **NFR-PERF-001:** The project brief's under-five-second analysis/outreach target is a later evaluation goal; it must be measured and reported under representative conditions before being claimed.
- **NFR-MAINT-001:** API request/response shapes and shared data fields shall be documented and agreed by backend and database owners before cross-module implementation.

8. Proposed acceptance criteria

Milestone 1 is acceptable when:

1. The agreed test set and results are recorded, with at least 95% completing the supported lead operations without an unhandled failure.
2. Invalid rows and duplicate imports are reported clearly.
3. Analysis is linked to the correct lead, persists, and can be retrieved.
4. All team members can follow the agreed local setup instructions.
5. The team can demonstrate the complete lead-to-saved-insight flow using approved sample data.
6. Open critical defects, their owners, and their disposition are included in the test/sign-off report.

9. Assumptions and decisions requiring team confirmation

Decision	Current baseline	Confirmation needed
Core stack	Python, FastAPI, Streamlit, Pydantic, SQLAlchemy, Alembic	Confirm exact dependency versions in the repository
Database	SQLite for local development; PostgreSQL for production	Confirm whether CI/tests must also run against PostgreSQL

Decision	Current baseline	Confirmation needed
AI provider	OpenAI API	Confirm model, quota, and key access; never place a real key in Git
Lead status values	New, Contacted, Qualified, Proposal, Won, Lost	Approve or replace
Duplicate rule	Normalized email; otherwise company + contact	Approve or replace
CRM integration	Not committed to a vendor	Select vendor and synchronization scope in a later milestone
Authentication	Excluded from initial local milestone	Required before production deployment
Repository	`https://github.com/230701224/sales-genie-ai.git` supplied by requester	Confirm access and the actual local source folder before repo work

10. Review and sign-off

Signing indicates agreement with the scope, requirements, and open-decision handling above. It does not claim that implementation or testing is complete.

Role	Name	Decision (Approve / Changes requested)	Date	Signature / approval reference
Project Lead				
Backend Developer				
Database/DevOps Engineer				
Full Team representative(s)				

11. Reference material

- AI Sales Intelligence Platform.pdf
- SalesGenie - Milestone 1 Plan.pdf
- SalesGenie - execution guide.pdf
- SalesGenie AI Intern training guide.pdf
- SalesGenie AI – Complete Installation-Guide.pdf